

# EMI: Flux & Faraday's Laws

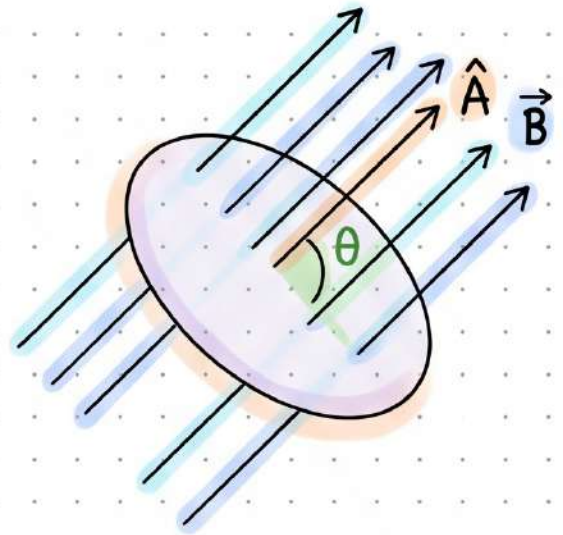
## Magnetic Flux ( $\phi$ ):

For a uniform magnetic field:

$$\phi = \vec{B} \cdot \vec{A} = BA \cos \theta \text{ weber.}$$

For a non-uniform magnetic field:

$$\phi = \int \vec{B} \cdot d\vec{A}.$$



## Faraday's Laws of EMI:

- (i) An induced emf is setup whenever the magnetic flux linking that circuit changes.
- (ii) The magnitude of the induced emf in any circuit is proportional to the rate of change of the magnetic flux linking the circuit,  $\epsilon \propto \frac{d\phi}{dt}$ .

## Lenz's Law:

The direction of an induced emf is always such as to oppose the cause producing it.

**Law of EMI:**  $\epsilon = -\frac{d\phi}{dt}$ . The negative sign indicated that the induced emf opposes the change of the flux.

## Motional EMF & AC Generators

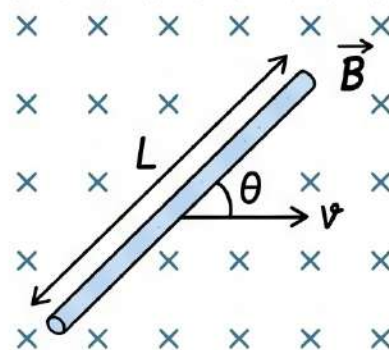
### Motional EMF

When a conductor is moved across a magnetic field, an electromotive force (emf) is produced in the conductor.

EMF Induced across a moving Straight Conductor in Uniform Magnetic Field:

$$E = BLv \sin \theta \quad \text{volt}$$

- $L$  = length of the conductor (m);
- $v$  = velocity of the conductor (m/s);
- $\theta$  = angle between direction of motion of conductor &  $B$ .

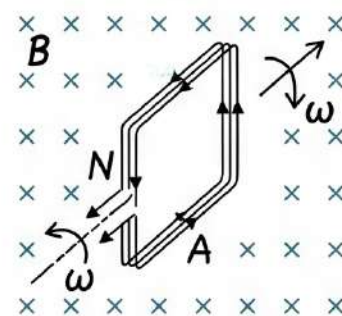


### Coil Rotation in Magnetic Field

Axis of Rotation is Perpendicular to the Magnetic Field.

Instantaneous induced emf:  $E = NAB\omega \sin \omega t = E_0 \sin \omega t$

- $N$  = number of turns in the coil
- $A$  = area of one turn
- $B$  = induction
- $\omega$  = uniform angular velocity of the coil
- $E_0$  = maximum induced emf.



### Super Conducting Loop in Magnetic Field

$R = 0$  ;  $\varepsilon = 0$  Therefore  $\phi_{\text{total}} = \text{constant}$

Thus through a superconducting loop flux never changes.



# Self & Mutual Induction

## Self Induction:

- The property of the coil or the circuit due to which it opposes any change of the current coil or the circuit is known as **Self-Inductance**. Its unit is **Henry**.

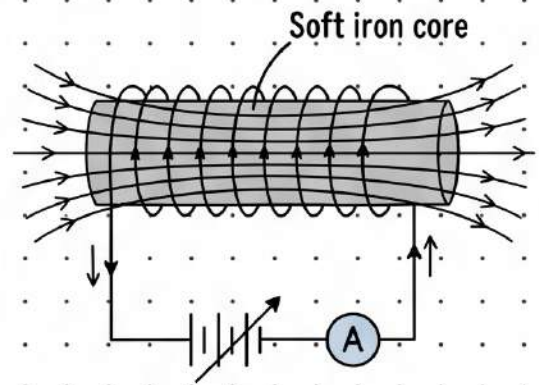
- Coefficient of Self inductance**

$$L = \frac{\phi_s}{i}, \text{ or } \phi_s = Li$$

- Self induced emf:  $e_s = -\frac{d\phi_s}{dt} = -L\left(\frac{di}{dt}\right)$   
(if  $L$  is constant)

- ⇒  $L$  depends only on:
- Shape of the loop and
  - Medium.

CETkit



Variable power coil

## Inductance of a Solenoid:

Magnetic field along the axis:

$$B = \mu ni \quad (\mu = \text{magnetic permeability; } n = \text{number of turns per unit length})$$

## Self inductance of a solenoid:

$$L = \mu n^2 A l \quad (A = \text{area of cross section})$$

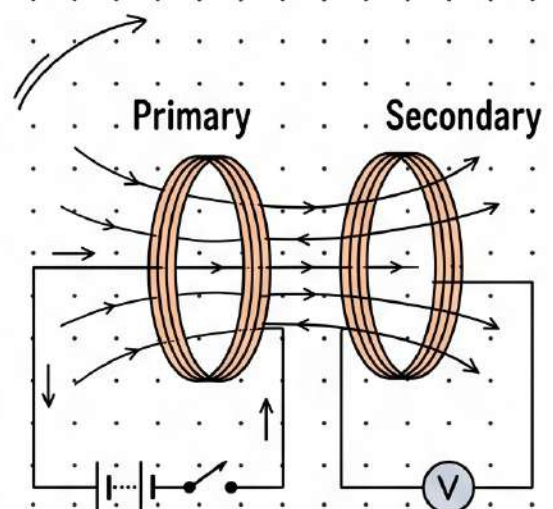
## Mutual Induction:

- Any change of current in one produces a change of magnetic flux in the other and the latter opposes the change by inducing an emf within itself.

- Mutual inductance =  $M = \frac{\text{flux linked with secondary}}{\text{current in the primary}}$

- Mutually induced emf ( $E_m$ ) =  $-M(dI/dt)$

- $M$  depends on
  - (1) geometry of loops
  - (2) medium
  - (3) orientation and distance between the loops.



# Energy & L-R Circuits

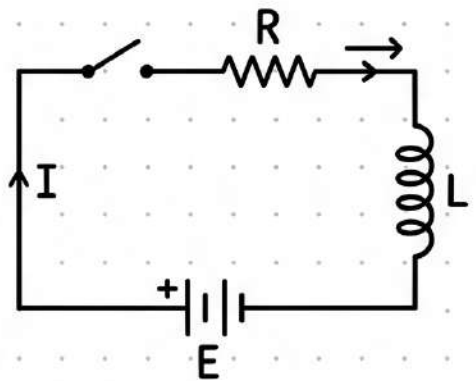
## • Energy in Inductors:

- **Energy Stored** in an **Inductor**:  $U = \frac{1}{2} LI^2$
- Energy of **interaction** of two **loops** =  $MI_1I_2$ , where  $M$  is mutual inductance.

## • Growth of a Current in an L-R Circuit:

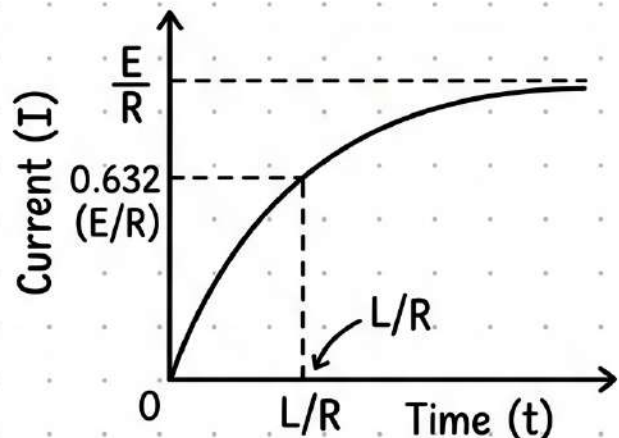
$$I = \frac{E}{R} (1 - e^{-Rt/L}) \quad (\text{If initial current} = 0)$$

$\frac{L}{R}$  = time constant of the circuit



- (i)  $L$  behaves as open circuit at  $t = 0$  ( $i = 0$ )
- (ii)  $L$  behaves as short circuit at  $t = \infty$

Growth of Current



## • Decay of Current:

- **Initial current** through the inductor =  $I_0$ .
- Current at any instant  $i = I_0 e^{-Rt/L}$